```
import cv2
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload image
uploaded = files.upload()

# Get image name
filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, 0)

# Sobel X-axis
sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)

# Sobel Y-axis
sobely = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

# Combine X and Y edges
edges = cv2.addWeighted(sobelx, 0.5, sobely, 0.5, 0)

# Convert to 8-bit image
edges = cv2.convertScaleAbs(edges)

# Save output
cv2.imwrite("Edge_detection.jpg", edges)

# Display original image
print("Original Image:")
cv2_imshow(img)

# Display output
print("Sobel XY Edge Detection:")
cv2_imshow(edges)

print("Output saved as Edge_detection.jpg")
```

Choose files img3.jpeg

img3.jpeg(image/jpeg) - 10583 bytes, last modified: 20/07/2026 - 100% done

```
Saving img3.jpeg to img3.jpeg
Original Image:
```

```
Sobel XY Edge Detection:
```

```
Output saved as Edge_detection.jpg
```